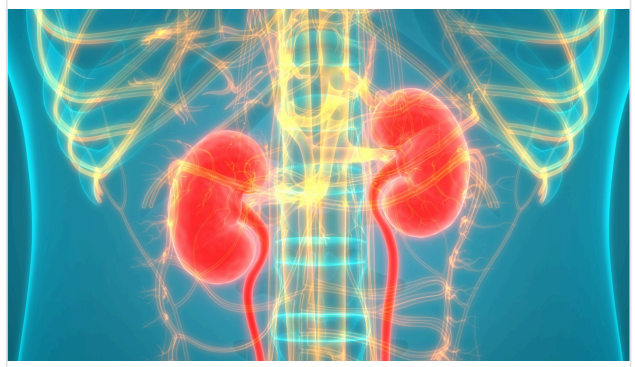
Chronic Kidney Disease Basics

MAY 15, 2024

For Everyone

KEY POINTS

- More than 1 in 7 American adults has chronic kidney disease (CKD), and as many as 9 in 10 don't know they have it.
- Specific blood and urine tests are needed to check for CKD.
- CKD can be treated (the earlier treatment starts the better).



MORE INFORMATION

For Everyone | For Health Care Providers | For Public Health

About your kidneys and CKD

Your hard-working kidneys

Your kidneys, each just the size of a computer mouse, filter all the blood in your body every 30 minutes. They work hard to remove wastes, toxins, and excess fluid. They also:

- Help control blood pressure.
- Signal the body to make red blood cells.
- Help keep your bones healthy.
- Regulate blood chemicals that are essential to life.

Kidneys that function properly are critical for maintaining good health.

CKD

CKD is a condition in which the kidneys are damaged and can't filter blood as well as they should. Because of this, excess fluid and waste remain in the body and may cause health problems such as heart disease.

Other health problems related to CKD include:

- Anemia or low number of red blood cells.
- Increased occurrence of infections.
- Low calcium levels, high potassium levels, and high phosphorus levels in the blood.
- Loss of appetite or eating less.
- Depression or lower quality of life.

CKD has varying levels of seriousness. It usually gets worse over time, though treatment has been shown to slow progression. CKD can progress to kidney failure or end-stage kidney disease (ESKD). It also increases the risk of heart disease and stroke.

When the kidneys stop working, dialysis or kidney transplant is needed for survival. Kidney failure treated with dialysis or kidney transplant is called end-stage kidney disease. Not all people with kidney disease progress to kidney failure.

Symptoms

People with CKD may not feel ill or notice any symptoms. The only way to find out for sure if you have CKD is through blood and urine tests. These tests measure both the creatinine level in the blood and protein in the urine.

Reducing risk

- Keep your blood pressure below 140/90 mm Hg (or the target your doctor sets for you).
- If you have diabetes, stay in your target blood sugar range as much as possible.
- Get active. Physical activity helps control blood pressure and blood sugar levels.
- Lose weight if needed.
- Get tested for CKD regularly if you're at risk.
- If you have CKD, meet with a dietician to create a kidney-healthy eating plan. The plan may need to change as you get older or if your health status changes.
- Take medicines as instructed and ask your doctor about blood pressure medicines called angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers, which may protect your kidneys in addition to lowering blood pressure.
- If you smoke, make a plan to quit. Smoking can worsen kidney disease and interfere with medication that lowers blood pressure.
- Include a kidney doctor (nephrologist) on your health care team.

KEEP READING
Risk Factors for Chronic Kidney Disease

CKD by the numbers

- Kidney diseases are a **leading cause of death** in the United States.
- About **35.5 million** US adults are estimated to have CKD, and most are undiagnosed.
- **40%** of people with severely reduced kidney function (not on dialysis) are not aware of having CKD.
- Every day, **360** people begin dialysis treatment for kidney failure.
- In the United States, diabetes and high blood pressure are the leading causes of kidney failure, accounting for **2 out of 3 new cases**.
- In 2022, treating Medicare beneficiaries with CKD cost **$95.7 billion**, and treating people for dialysis cost **$45.3 billion**.

KEEP READING
Chronic Kidney Disease: Common, Serious, and Costly

Resources

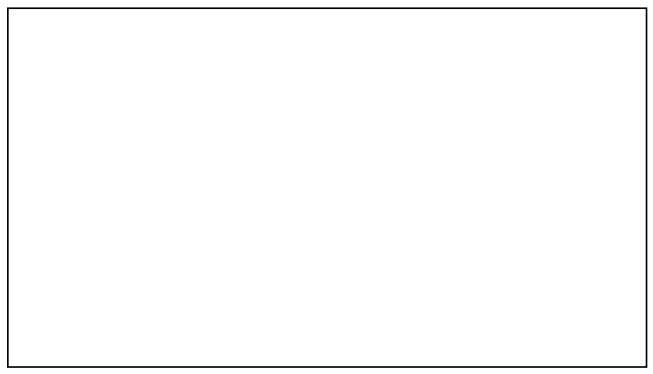
- Kidney Disease MP3 - Podcast (Running time 4 minutes 5 seconds)
- Native Americans, Diabetes, and Kidney Failure MP3 - Podcast (Running time 1 minute 14 seconds)

Videos

About Chronic Kidney Disease

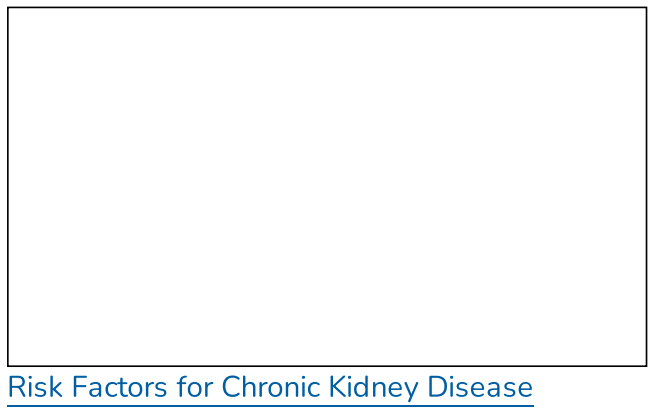
About Chronic Kidney Disease

Managing Chronic Kidney Disease



Managing Chronic Kidney Disease

Risk Factors for Chronic Kidney Disease



Risk Factors for Chronic Kidney Disease

Get email updates

Sign up here for CKD news and updates delivered directly to your inbox.

About the CKD Initiative

CDC's CKD Initiative collaborates with other government agencies, universities, and national organizations to:

- **Prevent and manage** risk factors for CKD.
- **Raise awareness** of CKD and its complications.
- **Promote early diagnosis and management** of CKD.
- **Improve outcomes** for people living with CKD.

The Kidney Disease Surveillance System documents CKD and its risk factors in the United States. The system tracks progress in preventing, detecting, and managing CKD. These efforts align with Healthy People objectives for CKD.

SOURCES

CONTENT SOURCE:
National Center for Chronic Disease Prevention and Health Promotion (NCCDPHP)